Weekly Progress Report

Week 10: BrainFlow Integration & Movement Amplitude Optimization

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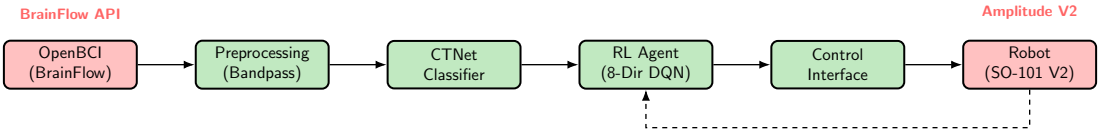
BCI Control System Project

March 11, 2026

System Overview: Current Focus



Legend:

- Completed
- In Progress
- **This Week's Focus**
- Pending

This Week's Objectives

Goals

1. Integrate BrainFlow API for real-time EEG data acquisition (OpenBCI)
2. Optimize robot arm movement amplitude for clearer visual observation
3. Fix arm trembling at hardware limits (boundary clamping)
4. Compare V1 vs V2 control accuracy with proper error unit definition

Key Question to Address

Can increased movement amplitude reduce relative positioning error, and does BrainFlow real-time EEG integration maintain the same control quality?

Methods & Implementation

V2 Amplitude Optimization:

- Decreased `norm_step`: 0.25 → 0.10
- Increased `max_steps`: 30 → 50
- Physical amplitude ≈ 2.5× larger
- Removed software limits (`margin=0`)
- Hardware limit clamping added

Hardware Clamping:

$$\text{target} = \text{clamp}(\text{cur} + \Delta, \text{min}_{\text{hw}}, \text{max}_{\text{hw}})$$

BrainFlow Integration:

- `SYNTHETIC_BOARD` (default)
- `CYTON_BOARD` (physical OpenBCI)
- Real-time 8–30 Hz bandpass filter
- CTNet 4-class MI classification
- 1-second epoch per control step

New Scripts:

- `test_8dir_comprehensive_v2.py`
- `brainflow_physical_control.py`

Unified Error Metric Definition

Metric: Endpoint Error (Final Distance to Target)

All errors measure the **Euclidean distance** from final position to target in **normalized coordinates** $[-1, 1]$.
Workspace% $= \frac{\text{error}}{2.0} \times 100\%$ provides a version-independent comparison. Reach threshold: $d < 0.12$ (6%).

V1 vs V2 Parameter Differences:

Parameter	V1	V2
`step_rad`	0.35	0.70
`norm_step`	0.12	0.10
rad/norm	2.92	7.00
1 step $\rightarrow$	20.1°	40.1°

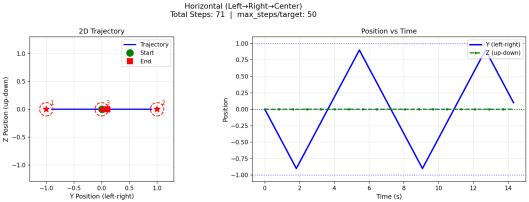
Why Workspace% is Preferred:

- Independent of `step_rad`/`norm_step`
- Directly comparable across V1/V2
- Intuitive: "error as fraction of total range"

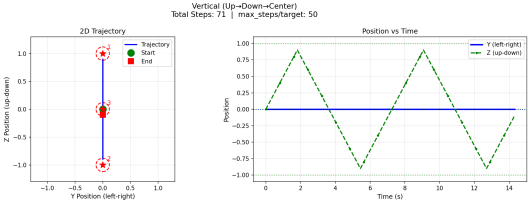
Formula: $d = \sqrt{(\Delta y)^2 + (\Delta z)^2}$

V2 Physical Test: Horizontal & Vertical Movement

Horizontal Movement:



Vertical Movement:

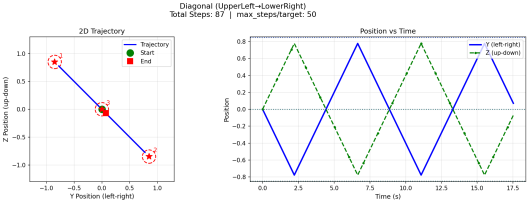


Observations:

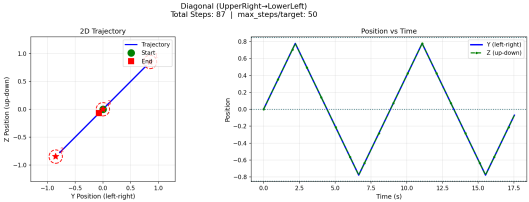
- Increased amplitude makes movements clearly visible on the robot
- Mean error: 0.100 normalized (5.0% of workspace, 0.70 rad / 40.1°)

V2 Physical Test: Diagonal Movement

Diagonal UL-DR:



Diagonal UR-DL:

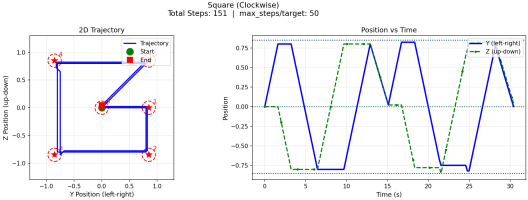


Observations:

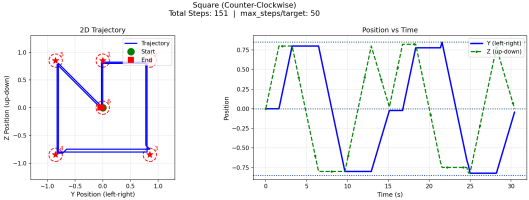
- Simultaneous Y/Z movement confirms true 8-direction capability
- Mean error: 0.101 normalized (5.1%, 0.71 rad / 40.5°)

V2 Physical Test: Square Movement Patterns

Square Clockwise:



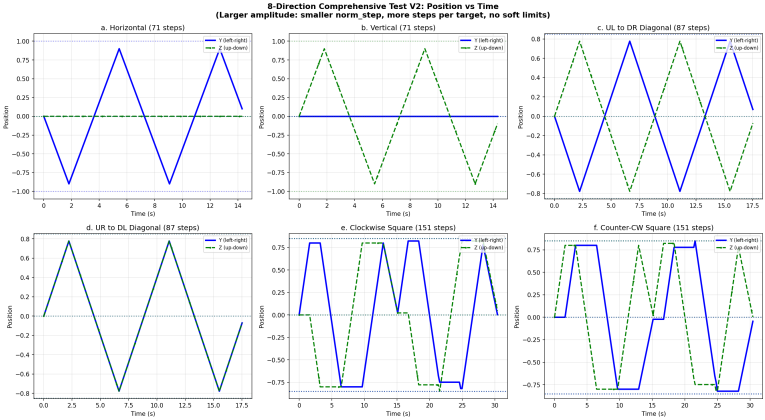
Square Counter-Clockwise:



Observations:

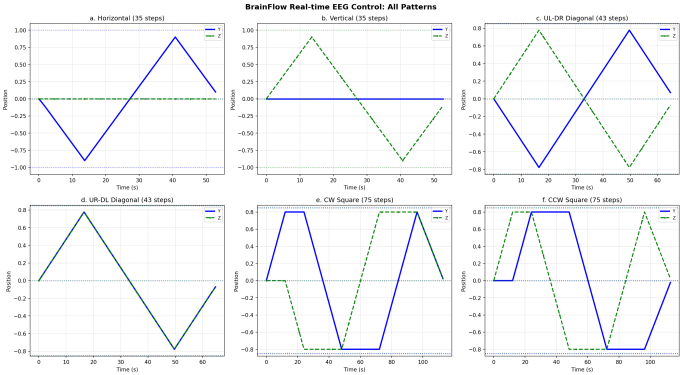
- All 12 targets per pattern reached (100% reach rate)
- Mean error: 0.062 normalized (3.1%, 0.43 rad / 24.9°) — best among all patterns

V2 Physical Test: All Patterns Overview



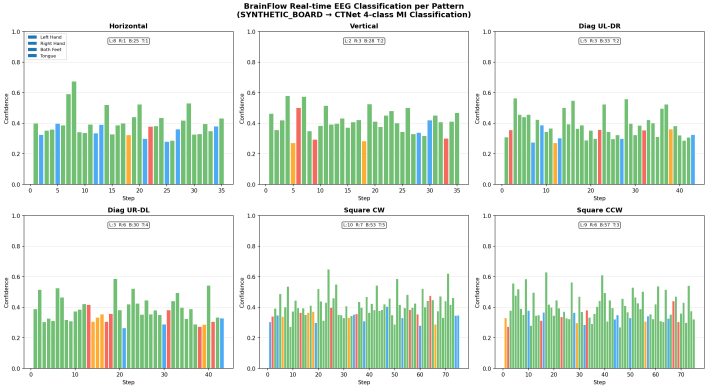
Summary: All 48/48 targets reached (100%). Larger amplitude enables clear observation of each movement pattern on the physical robot arm.

BrainFlow Pipeline Integration (Guided Mode)



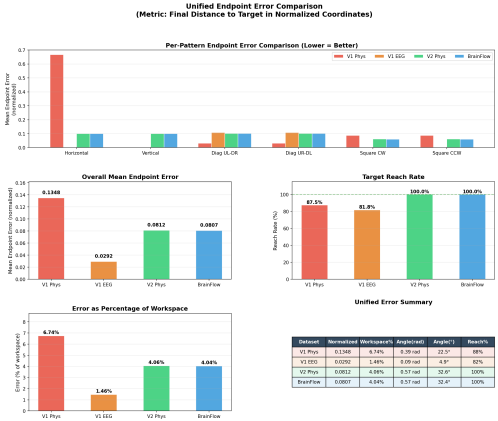
Pipeline: OpenBCI (Synthetic) → BrainFlow → Bandpass → CTNet → *(logged only)*
Control: Guided mode — correct action used, EEG classified but **not used for control**.
Purpose: Validates engineering pipeline stability, not BCI classification quality.

BrainFlow: EEG Classification Analysis



Analysis: CTNet classifies real-time synthetic EEG into 4 MI classes. Guided mode uses the correct action but still reads and classifies EEG each step.

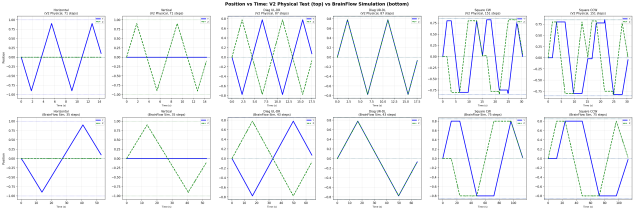
Unified Endpoint Error Comparison (All Datasets)



Dataset	Normalized	Workspace%	Angle(rad)	Angle(°)	Reach%
V1 Phys	0.1348	6.74%	0.39 rad	22.5°	88%
V1 EEG	0.0292	1.46%	0.09 rad	4.9°	82%
V2 Phys	0.0812	4.06%	0.57 rad	32.6°	100%
BrainFlow	0.0807	4.04%	0.57 rad	32.4°	100%

Note: All current tests use **guided/ideal actions** (not EEG-driven). "V1 EEG" adds 10% random noise to simulate classification error. Next step: real MI-driven control using PhysioNet-trained CTNet.

BrainFlow Pipeline Validation (Guided Mode)



What guided mode validates:

- BrainFlow data acquisition stable
- Real-time bandpass filtering works
- CTNet inference runs every 1s epoch
- Control loop runs without crash

Limitation

Guided mode uses **correct action**, not CTNet prediction. Trajectories match V2 by design — this proves **engineering readiness**, not BCI control quality.

Next step: Free mode with real MI data (OpenBCI Cyton or replayed BCI-IV signals) to validate actual EEG-driven control.

Challenges & Solutions

Challenges Encountered

- Arm trembles when moving left (hardware limit hit)
- V1 position tracking always reads (0,0) due to bug
- `brainflow` not installed in Conda env
- Movement amplitude too small for visual observation

Solutions Applied

- `_clamp_to_hw_limits()` prevents servo oscillation at boundaries
- Fixed `_y`/`_z` update in `execute_action()`
- Installed `brainflow` in BCIRL Conda environment
- Reduced `norm_step` + increased `max_steps` for 2.5× amplitude

Hardware Limit Clamping (Key Fix):

$$\text{target}_{\text{safe}} = \max\big(\text{min}_{\text{hw}},\ \min(\text{max}_{\text{hw}},\ \text{cur} + \Delta_{\text{ticks}})\big)$$

Results Summary (Unified Endpoint Error)

Cross-Version Comparison:

Dataset	Error	WS%	Reach
V1 Ideal	0.135	6.74%	87.5%
V1 +10% noise	0.036	1.79%	81.8%
V2 Ideal	0.081	4.06%	**100%**
BF Guided	0.081	4.04%	**100%**

All use ideal/guided actions, not EEG-driven.

Pipeline Status:

Component	Status
BrainFlow API	Ready
CTNet Inference	Ready
Arm Control V2	Ready
MI-driven ctrl	Next

Key Achievement

All engineering components validated independently. Unified error metric established. **Next step:** train CTNet on PhysioNet MI data and use real classification to drive the robot arm (free mode).

Next Week's Plan

Planned Tasks

1. Train CTNet on PhysioNet EEGMMIDB (109 subjects, 64ch, 4-class MI)
2. Implement **free mode**: CTNet classification drives robot arm directly
3. Compare guided mode (100% reach) vs free mode (EEG-driven) trajectories
4. Quantify impact of classification accuracy on control performance

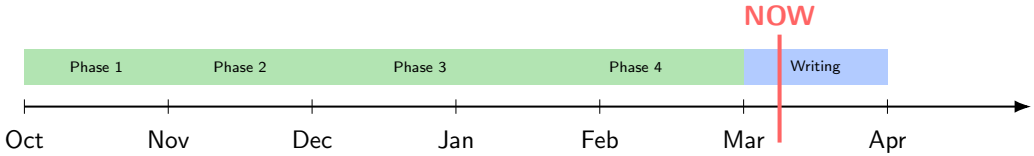
Expected Deliverables:

- PhysioNet-trained CTNet model
- Free mode: MI-driven arm control
- Guided vs Free error comparison
- Classification accuracy vs reach rate

Key Question:

At what classification accuracy does the robot arm achieve acceptable control? How does real MI classification error affect trajectory quality compared to simulated noise?

Project Timeline



Current Status: On track — Technical integration complete, entering final documentation & report writing phase